

CODE GUIDELINES AND LAYOUT RULES

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Project Directory Structure

To maintain code clean and readable, files are structured into functional folders:

- Root: Entry points app.py, train_models.py, generate_datasets.py.
- datasets/: Raw agricultural databases and logged user history.
- models/: Saved pickle (.pkl) files (scalers, encoders, estimators).
- templates/: HTML view pages (server, predict, dashboard, results).
- static/: CSS styling and generated performance visualizations.
- tests/: Unit test scripts.
- tests/test_app.py - unittest suite (4 tests)
- models/ - serialised ML pipelines (.pkl)
- templates/ - Jinja2 HTML pages
- static/ - CSS images

2. Naming and Formatting Standards

Python modules follow PEP-8 conventions. Methods use clear names (e.g. 'load_all_models()', 'train_and_compare()'), and variables include clear descriptions. Models are saved as pipelines to preserve the execution order of standardizers and encoders.

Conventions: PEP 8 formatting, snake_case naming, full docstrings, type hints where applicable.